

VAC PRESENTATION

❑ **Group Of B.sc(h) Mathematics**

❑ **PRESENTED TO :**

MR. GAGANDEEP LONGIANY SIR

TOPIC COVERED BY :

☐ **SHAILESH**

UNIVERSITY ROLL NO : 24035563026

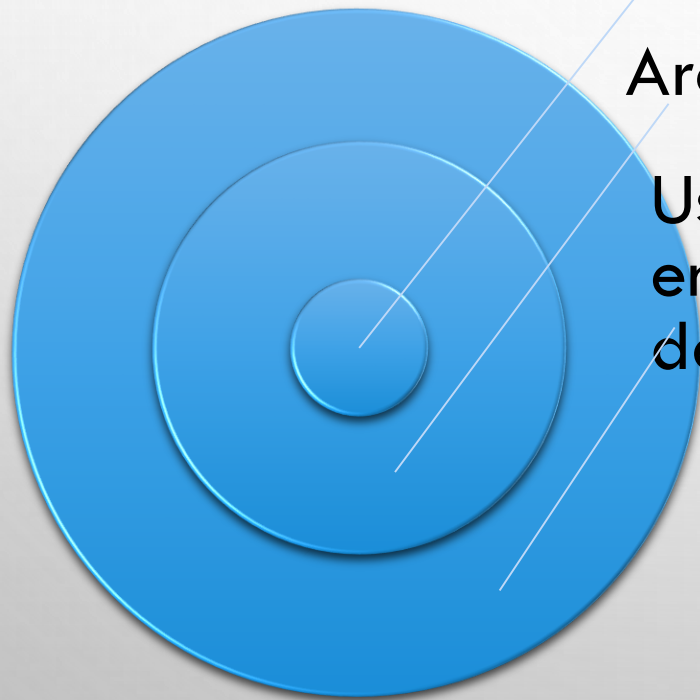
COLLEGE ROLL NO : 24/BS(H)MT/26145

☐ **SHIVAM YADAV**

UNIVERSITY ROLL NO : 24035563025

COLLEGE ROLL NO : 24/BS(H)MT/26149

INDEX :



Construction

Architecture and Design

Use of natural
environmental friendly
designs & materials



➤ INTRODUCTION :

- Modern construction has a major impact on the environment
 - Traditional methods consume high energy and natural resources
 - Sustainable architecture focuses on eco-friendly design and materials
 - Goal: reduce environmental damage while improving human comfort
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➤ WHAT IS ECO-FRIENDLY ARCHITECTURE ?

- Design approach that minimizes negative environmental impact
- Uses natural resources efficiently
- Focuses on long-term environmental balance
- Promotes healthier living and working spaces



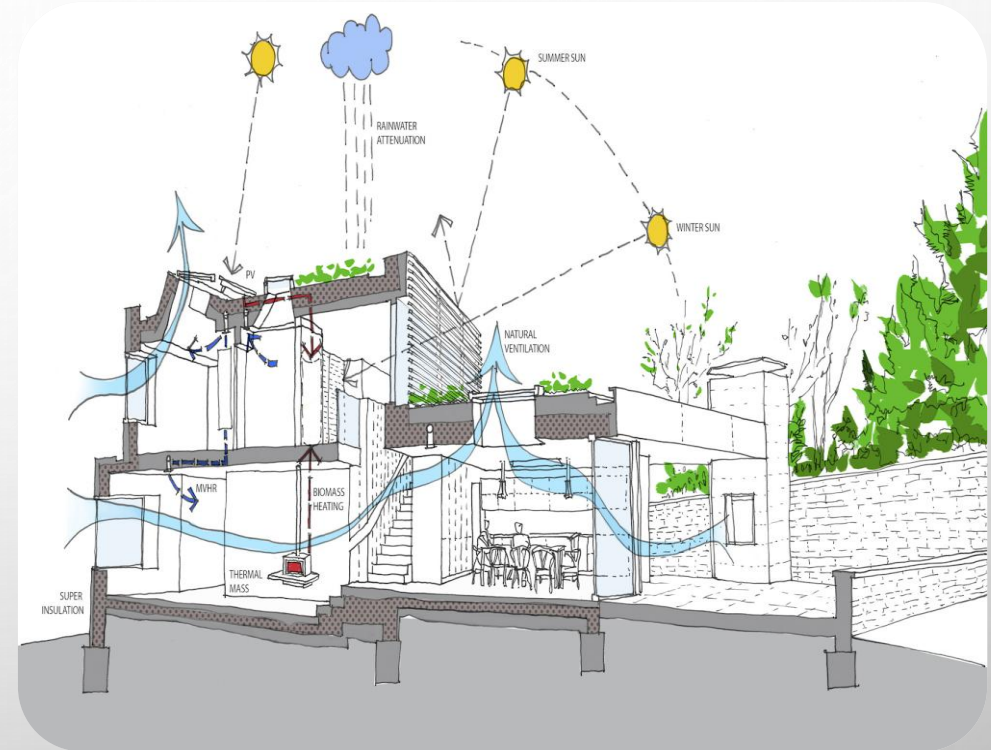
➤ IMPORTANCE OF ENVIRONMENT-FRIENDLY CONSTRUCTION

- Reduces carbon footprint
- Conserves natural resources
- Saves energy and water
- Improves indoor air quality
- Supports sustainable development



➤ PRINCIPLES OF NATURAL & ECO-FRIENDLY DESIGN

- Energy efficiency (natural lighting and ventilation)
- Use of renewable resources
- Climate-responsive design
- Waste reduction and recycling
- Harmony with natural surroundings



➤ USE OF NATURAL MATERIALS IN CONSTRUCTION

- **Wood** – renewable, strong, and flexible
- **Bamboo** – fast-growing and highly sustainable
- **Stone** – durable and naturally insulating
- **Clay & Mud** – excellent thermal regulation
- **Lime** – eco-friendly alternative to cement



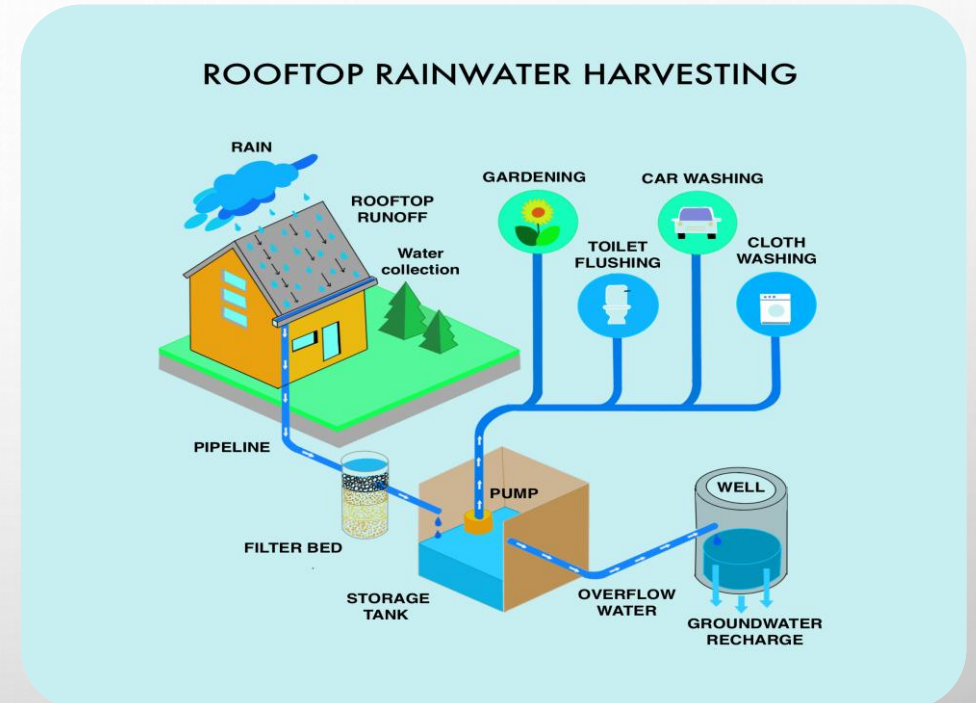
➤ ENVIRONMENT-FRIENDLY BUILDING MATERIALS

- Recycled steel and glass
- Fly ash bricks
- Rammed earth
- Cork and natural fibers
- Low-voc paints and finishes



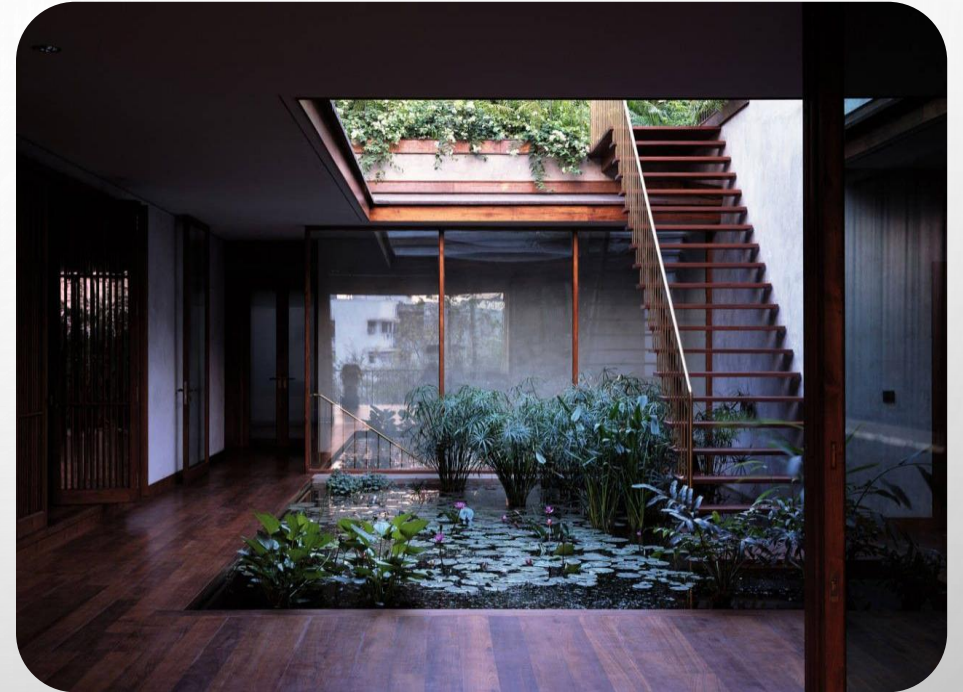
➤ ECO-FRIENDLY ARCHITECTURAL DESIGN TECHNIQUES

- Passive solar design
- Green roofs and walls
- Rainwater harvesting systems
- Natural cross-ventilation
- Orientation of buildings to reduce heat gain



➤ **ROLE OF NATURE IN ARCHITECTURAL DESIGN**

- Use of indoor plants and courtyards
- Integration of landscapes with buildings
- Daylight optimization
- Use of water elements for cooling effect



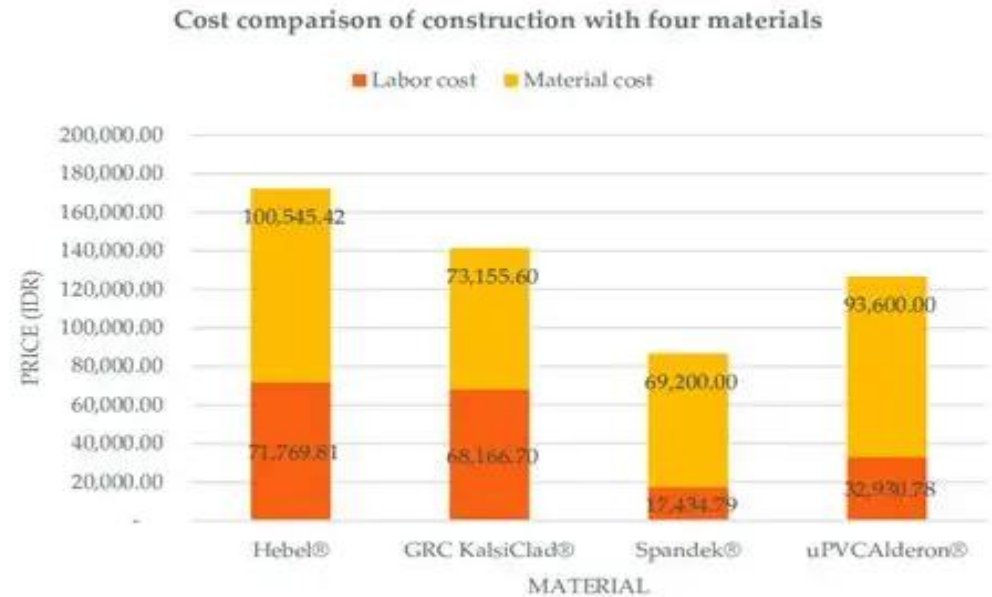
➤ BENEFITS OF SUSTAINABLE CONSTRUCTION

- Lower energy and maintenance costs
- Healthier indoor environment
- Increased building lifespan
- Reduced environmental pollution
- Better quality of life



➤ CHALLENGES IN ECO-FRIENDLY CONSTRUCTION

- Higher initial cost
- Lack of awareness
- Limited availability of materials in some areas
- Need for skilled professionals



➤ FUTURE OF SUSTAINABLE ARCHITECTURE

- Smart green buildings
- Increased use of renewable energy
- Stronger environmental regulations
- Greater public awareness
- Integration of technology with nature





➤ CONCLUSION

- Eco-friendly construction is essential for environmental protection
 - Natural materials and sustainable designs reduce ecological damage
 - Sustainable architecture ensures a better future for coming generations
 - Balance between development and nature is the key to progress
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- **Q.1** : What are the key principles of sustainable development, and why is it important for addressing global environmental challenges ? Provide three examples of sustainable practices in different sectors.

(Sr. No. of Q. Paper : 1841)

Answer : Sustainable development means development that meets the needs of the present generation without harming the ability of future generations to meet their needs. It focuses on balancing economic growth, environmental protection, and social welfare.

Key principles :

(1) Intergenerational equity:

- (i) we should use natural resources carefully so that future generations can also use them.
- (ii) all people in the present generation should get equal access to resources like water, food, and energy.

(2) conservation of nature – we must protect forests, wildlife, and biodiversity.

(3) Precautionary principle – If any activity can harm the environment, we should take preventive steps even if there is no complete scientific proof.

(4) Polluter pays principle – the person or industry that causes pollution should pay for the damage.

Importance:

(i) Helps reduce climate change.

(ii) Prevents overuse of natural resources and reduces pollution.

(iii) Protects plants and animals.

(4) supports long-term economic and social development.

Importance in addressing global environmental challenges:

(i) Helps combat climate change by promoting renewable energy.

(ii) Prevents resource depletion through sustainable use of natural resources.

- (i) Reduces **pollution** via cleaner production methods.
- (ii) Protects **biodiversity** and maintains ecological stability.
- (iii) Promotes long-term economic and social development.

Examples of sustainable practices:

- (i) **Agriculture** – organic farming and drip irrigation.
- (ii) **Energy** – use of solar and wind energy instead of coal and petrol.
- (iii) **Industry** – recycling waste and using energy-saving machines.

Conclusion:

Sustainable development is important for protecting the environment and ensuring a better future for the next generations. It helps in solving global environmental problems while supporting economic growth.

■ Q.2 : Agricultural practices from ancient to modern times.

(Sr. No. of Q. Paper : 1841)

Answer :

Agriculture in Ancient times:

- (i) In ancient times, agriculture was mainly subsistence farming, where farmers grew crops for their own consumption.
- (ii) Simple tools like wooden ploughs and sickles were used.
- (iii) Farming depended on rainfall.
- (iv) Natural manure such as cow dung and compost was used.
- (v) Shifting cultivation was practiced in some areas.

Agriculture in Medieval period:

- (i) Iron ploughs were introduced.

(li) irrigation improved through wells, tanks, and canals.

(lii) crop rotation and mixed farming were practiced.

Agriculture in modern times : Modern agriculture developed mainly after the **green revolution** in the 1960s.

(i) Use of high yielding variety seeds.

(ii) Chemical fertilizers and pesticides.

(iii) Tractors, harvesters, and machines.

(iv) Large-scale commercial farming.

Conclusion:

Agriculture has moved from simple traditional methods to modern technology-based methods. While modern agriculture increases food production, sustainable farming is important to protect the environment for future generations.

Thank You!